

Legislative Information Resources Management for House Bills and House Resolutions in the Acquisition Department of the House of Representatives

CHRISTALYN B. LAPID

Information Systems Department
College of Information and
Computing Sciences,
University of Santo Tomas
España, Manila City Philippines
christalyn.lapid.iics@ust.edu.ph

DIANNE NICOLE M. SILVA
Information Systems Department
Institute of Information and
Computing Sciences,
University of Santo Tomas
España, Manila City Philippines
diannenicole.silva.iics@ust.edu.ph

ALYANNA DAWN S. MANUEL

Information Systems Department
College of Information and
Computing Sciences,
University of Santo Tomas
España, Manila City Philippines
alyannadawne.manuel.iics@ust.edu.ph

JOERJI LOUIS A. ONG

Information Systems Department
Institute of Information and
Computing Sciences,
University of Santo Tomas
España, Manila City Philippines
joerji.louis.ong.iics@ust.edu.ph

ARNE B. BARCELO

Information Systems Department
Institute of Information and
Computing Sciences,
University of Santo Tomas
España, Manila City Philippines
abbarcelo@ust.edu.ph

Abstract—The government typically manages a lot of documents during their day to day operations. Aside from documents, it also requires a large amount of effort in monitoring the compliance of each employee in the government. Given the current problem, the researchers provided Configuration Management and Monitoring and Event Management [20] Information Technology Infrastructure Library (ITIL) frameworks to provide a systematic way of receiving, archiving, and transmitting vital government documents which are the House Bills and House Resolutions. With that, the researchers created the LIRM: Legislative Information Resources Management for House Bills and House Resolutions in the Acquisition Department of the House of Representatives. The development of the system was done through the Agile methodology. This methodology was used for the purpose of saving time and acquiring continuous feedback and results during each sprint of the development. The local-based application was developed using PHP, Angular and other technologies that assures the system's performance, reliability, availability, and security. Tests performed were based on the [16] Institute of Electronics and Electrical Engineering (IEEE) standard for Software Test Documentation to make sure that the standards and acceptance criteria are met.

Keywords—*Legislative Information Resource Management; House Bills; House Resolutions; ITIL framework; Configuration Management; Monitoring and Event Management; Service Management.*

I. INTRODUCTION

In today's world, technology has been an integral part to the progress of government organizations. IT modernization is a key component of one's productivity when it comes to improving business processes in government organizations.

Documents are important to any government as it is where information essential to the public is kept thus it provides a trusted source of information. When documents are organized poorly, problems might occur if there are needed documents to be searched and may require extensive time and effort that varies depending on the number of files and documents cluttered.

The Acquisition Department of the House of Representatives aims to reform the conventional system of document and compliance process to address the problems in the current system. The study provides a solution wherein the House Bills and House Resolutions will be secured, highly available and the compliance of every employee will be monitored efficiently. The researchers provided a way to improve the management of House Bills and House Resolutions of the House of Representatives that can provide reference and information services through precise management and tracking mechanism of documents.

IT Service Management enables the government to maximize their potential by utilizing the appropriate technology. ITIL framework that the study targeted were the configuration management and monitoring and event management. In this era of technology, the acquisition department of the House of Representative has a vital role to keep up with their important records wherein the technology is maximized within the system created. This is also for the

continuous improvement of their services for the legislative branch, which in turn serve the citizens of this country through the records it organized and the service it improved. This assumes full responsibility in the appraisal, selection and acquisition of records for safekeeping and preservation. The LIRM would consist of the following users:

a. *System Administrator*

Researchers are the system administrator to be able to oversee and maintain the application software. In the long run, an IT staff member from the Acquisition Department will be acting as the System Administrator to manage and monitor the LIRM.

b. *Supervisor*

Manages and monitors the legislative documents in the LIRM. They can also encode House Bills and House Resolutions.

c. *Director*

Oversee the LIRM, without having any access in updating the documents. Director still must log in.

d. *Staff*

Responsible for encoding the details of the House Bills and House Resolutions to the LIRM.

e. *Guest*

Congress staff and house secretariat that can view active and archived House Bills and House Resolutions without the need to log in.

A. OBJECTIVES

The purpose of the study led to the development of Legislative Information Resource Management (LIRM) that manages the House Bills and House Resolutions in the Acquisition Department of the House of Representatives.

To attain the objectives of the study, the following are the specific objectives:

1. To develop an application software that will be utilized by the Acquisition Department of the House of Representatives.
2. To develop an application software that will improve the management of the day-to-day receiving,

transmitting and archiving processes of House Bills and House Resolutions.

3. To implement the Information Technology Infrastructure Library version 4 (ITIL v4) components such as Configuration Management and Monitoring and Event Management in developing the application software.

B. ITIL FRAMEWORK

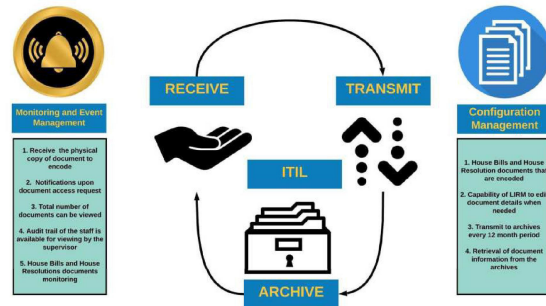


Fig. 1. ITIL Framework

Monitoring and Event Management

In Legislative Information Resource Management (LIRM), monitoring takes place whenever a user can keep track of the activities to or from a user. To be precise, the administrator would have all out access and specifically visibility of all documents that are inputted and as well as know the activities being done for all users of the LIRM. With an exemption of having less privileges of the staff members, they have access to their own activities. An event is considered whenever one of these activities is done successfully. The privilege to monitor all activities by the administrator and the capability of the staff to know their own progress works together with multiple events occurring at the same time.

Configuration Management

Configuration Management could be found in the LIRM whenever the Configuration Items (CI) which are the House Bills and House Resolutions are being created, added, or modified. The CI are the components that are being managed to execute a service. The CI is being handled and managed by the staff members and if needed the administrator also has the power to help with the input of CI to the LIRM. The CI has their own description once they are inputted in the LIRM and may be edited if there may be human error involved or updated and deleted if there is a need.

C. PROPOSED BUSINESS PROCESS

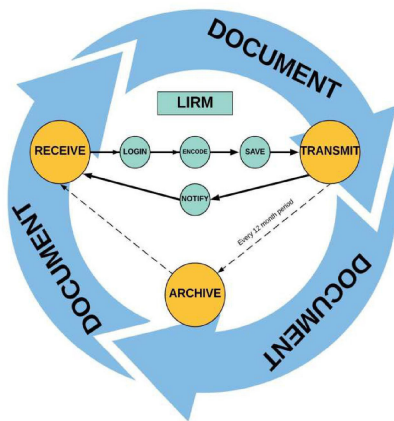


Fig. 2. Proposed Business Process

Fig. 2. Represents the conceptual framework of the House Bills and House Resolutions around the Legislative Information Resource Management (LIRM). Through the three main tasks which are receiving, transmitting, and archiving.

The Acquisition department encodes the received House Bills and House Resolutions documents. When the encoding was done, the staff will then transmit the encoded records to the supervisor. The supervisor will then be notified if a staff member transmitted a certain record for monitoring. After each year, the encoded House Bills and House Resolutions will be archived for future use. This is all done within the LIRM. Hence, a full cycle has been created

II. METHODOLOGY

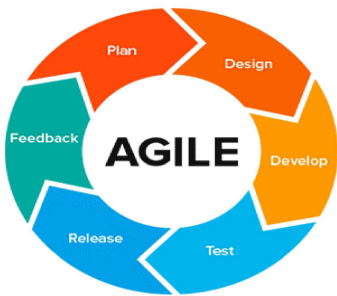


Fig. 3. Agile Methodology

The researchers used the Agile methodology for implementing the application software. Fig. 3. above shows the development process of the Legislative Resource Management (LIRM). It consists of Planning, Designing, Developing, Testing, Releasing, and Feedback. The planning stage consists of scheduling of internal meetings as well as with the Client, gathering of requirements and risks, taking up available tasks,

understanding the difficulty and effort that is needed to accomplish tasks, and the like. Next, the designing and developing of the application with the use of mockups as the guide in programming the front end and the back end of the project. The testing stage is when the finished features would have a series of tests for quality assurance checking. Finally, when all tasks are completed and all test cases have been satisfied, there would be the releasing stage, in which the application would be deployed in the Client's location.

III. RESULTS AND DISCUSSION

The testing phase of this project was reconstructed from the [16] Institute of Electrical and Electronics Engineers or the IEEE. The IEEE 829-1998 format was a software test documentation that defines the format and contents of individual test documents dynamically. Overall, it includes the introduction, test items, features to be tested, test approach, item pass/ fail criteria, suspension/resumption requirements, deliverables, testing tasks, approvals, and responsibilities. The purpose of the testing standard is to meet the [16] IEEE standard. This is to ensure the consistency in practice and consistency in the results with regards to the international standards.

The researchers had tested the following functionalities of the system and ensured that the acceptance criteria were met. Using the [16] IEEE standard of testing, each functional and nonfunctional requirement yields a "pass" status as shown in Tables 1, 2, and 3.

A. Functional Requirements

Requirement Statement	Output	Status
The system shall have a database that will store, display, and list the House Bills and House Resolutions	<i>MySQL</i>	PASS
The system shall have a functionality where employees are able to encode the House Bills and House Resolutions	<i>Add Documents</i>	PASS
The system shall have a functionality where employees can see a visual graph of all current House Bill and House Resolutions	<i>Dashboard</i>	PASS

The system shall have a notification feature which serves as an indicator for system actions	<i>Notification bell</i>	PASS
The system shall have a functionality where employees can archive House Bills and House Resolutions that would not be used anymore but would be kept for future references.	<i>Archive</i>	PASS
The system shall have a functionality where the system administrator can approve, disapprove newly created users.	<i>Configuration of users</i>	PASS
The system shall have a functionality where employees outside the House of Representatives cannot access the system due to its security configurations	<i>Security configuration</i>	PASS

B. Non-Functional Requirements

Requirement Statement	Output	Status
The Legislative Information Resource Management requires a LAN connection to access the application software. The users can access the application software by using a browser.	<i>Technical Environment / Network Topology</i>	PASS
Since the application software created is web-based, a desktop or a laptop is required with a browser to run the	<i>Hardware Requirements</i>	PASS

application software. The required device/s should have LAN/WLAN connection and browser installed to the device/s. The memory needed would be 300 MB RAM and a storage of at least 1 GB disk space. The operating systems that will be needed are Windows XP, Vista, 7, 8 or later, or MAC OS.		
The application software should not encounter any problems running with operating systems such as Windows and Mac OS. The following tools are needed to accomplish the application software: - Google Angular (HTML, CSS) - Web browser (Preferably Google Chrome) - MySQL - PHP - Visual Studio	<i>Software Requirements</i>	PASS
To maximize the performance of the application software, it must be able to: - Be accessible anytime within the vicinity of the Acquisition Department. - Researchers performed unit testing and user acceptance testing to ensure that the application software is performed as intended. - Compatible with the specifications of the computers. No need to rely on fast internet connection.	<i>Performance Requirements</i>	PASS
System administrators of supervisors can always edit the document settings to show the appropriate	<i>Reliability Requirements</i>	PASS

details for the incoming congress periods. To ensure the system's reliability, the following conditions must be considered: - Since it uses a LAN with the ITIL components, it is accessible only within the Acquisition - Department wherein the System Administrator would run the needed servers for the Application to run together with the other PCs. - The new user accounts functionality of the application can only be accepted/declined by the System Admin and Supervisor for them to thoroughly check the credibility of the new users. - The server would have enough space for the database for information		
Ceaseless support must be under the responsibility of the System Administrator. In any occurrence of loss information, the System administrator will be held liable. In case of maintenance, every new congress period, the House Bills/Resolutions will be archived by the IT administrator or supervisor.	<i>Supportability Requirements</i>	PASS
The overall scope and application functionality are documented by using different accounts which have different access levels to the application.	<i>User Documentation Requirements</i>	PASS

The application software will have different levels of access in terms of their account to ensure the validity of data to prevent future losses and damages. The supervisor and system administrator confirms new account requests for new staff or users. If ever there are any instances of lost username or password, the staff, supervisor, and director must have to contact the administrator to manage their account. There will be five different security questions. They will choose one of these questions through drop down and it must be answered for them to gain access to their accounts. User can only access the application software on one device only.	1. <i>Security and Privacy Requirements</i> 2.	PASS
---	---	-------------

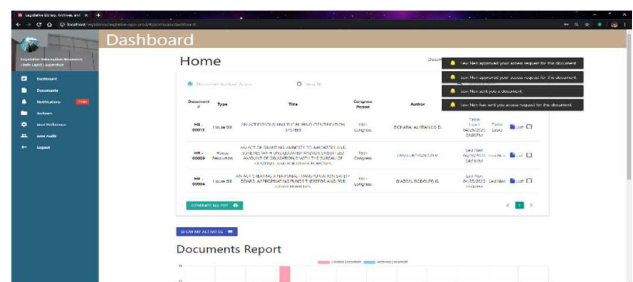


Fig. 4. Dashboard for monitoring House Bills and House Resolutions

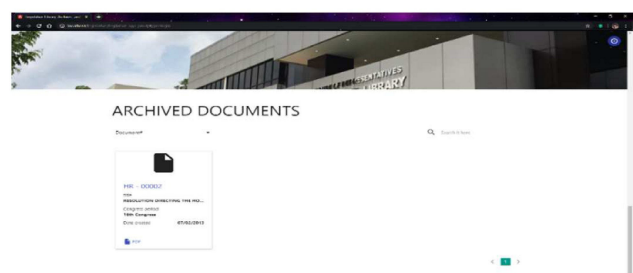


Fig. 5. List of Archived documents



Fig. 6. Visual graph of House Bills and House Resolutions

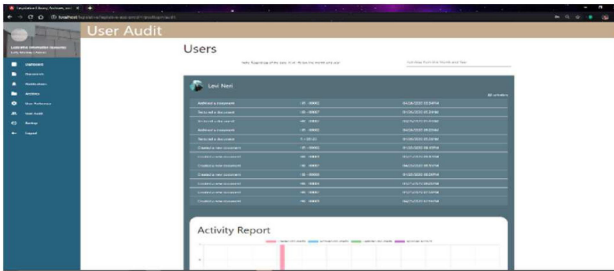


Fig. 7. Configuration of Users

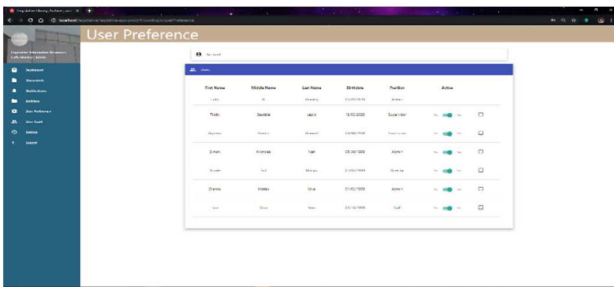


Fig. 8. Configuration of Users

IV. CONCLUSION AND RECOMMENDATION

The researchers were able to develop the Legislative Information Resources Management (LIRM) of House Bills and House Resolutions integrating ITIL v4 components (Monitoring and Event Management, and Configuration Management). User acceptance tests (UAT) were also done by the researchers, where the positive feedback was mostly received with minimal recommendations focusing on the user interface. The functionalities of the system are working as expected. The recommendations of the researchers are to include other legislative documents of the Philippine Government to centralize the storage location of all documents as there are only two types of documents are currently being inputted in the LIRM, also have an automated document encoder to make the staff members work more efficient and lessen the processes involved in encoding a Configuration Item, and implement a ticketing system for guests so that there will

be an efficient way to get new updates of new documents entering the LIRM versus having a personal messenger delivering documents to the premises of the Client.

V. REFERENCES

- [1] Asadi, S. (2011). Logistics System. *Logistics Operations and Management*, pp.221-245.
- [2] Backman, K. F. (2018). Event management research: The focus today and in the future. *Tourism Management Perspectives*, 25, pp 169–171.
- [3] Chen, W.-M. & Chen, Y.-C., 2012. Web design and implementation for remote control. In *Intelligent Control and Automation (WCICA)*. Beijing, 2012. IEEE.
- [4] Coronel, Carlos, and Morris, Steven. *Database Systems: Design, Implementation, and Management*. 2015 Cengage Learning, 2017
- [5] David, M. (2010). *HTML5*. 1st ed. Hoboken: Taylor and Francis, p.3.
- [6] Dr. Gregory S. Dawson, A Roadmap for IT Modernization in Government, (Washington, D.C., IBM Center for The Business of Government, 2018).
- [7] Farrell, J. (2012). *Java programming*. 6th ed. Australia: Course Technology Cengage Learning, p.40.
- [8] Gabarro, S., 2007. The Different Approaches of Web Programming. In *Web Application Design and Implementation: Apache 2, PHP5, MySQL, JavaScript, and Linux/UNIX*. 1st ed. Wiley-IEEE Press. pp.17 - 26.
- [9] Hope, A. and Forrest, G. (2010). Databases. *Anaesthesia & Intensive Care Medicine*, 11(12), pp.495-496.
- [10] IBM.com. (2019). *What is IT Service Management (ITSM)?* Retrieved from <https://www.ibm.com/cloud/learn/it-service-management#toc-what-is-it-XWLLNALy> [Accessed 18 Sep. 2019].
- [11] Information Resource Management. (n.d.) *The Free On-line Dictionary of Computing*. (2003). Retrieved December 9 2019
- [12] ITIL®: the basics Valerie Arraj, Compliance Process Partners LLC. (2013). 3rd ed. [ebook] p.3. Retrieved from <https://www.mountainview-itsm.com/itil-training/downloads/axelos/8863-WP-ITIL-the-basics.pdf> [Accessed 25 Nov. 2019].
- [13] Khosravi, S. and Afshari, H. (2010). Information resources management: An implementation methodology. The 40th International Conference on Computers & Industrial Engineering.
- [14] Lester, E. (2017). Configuration Management. *Project Management, Planning and Control*, pp.105-106.
- [15] Mrantinkovic, Aleksander. *Theory, Practice and Techniques in HTML and CSS*. 3G E-
- [16] 829-1983 - IEEE Standard for Software Test Documentation (1991, Mar 21). IEEE Standard | IEEE Xplore. <https://ieeexplore.ieee.org/document/573169>